

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

MEDICAL ELECTRONICS

(Maximum marks: 100)

PART-A

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. What are microelectrodes?
2. List the composition of blood.
3. State the need of defibrillators.
4. List the different types of diathermy equipments.
5. State any two applications of NMR imaging. (5*2=10)

PART-B

II. Answer any five of the following questions. Each question carries six marks.

1. Explain the electrical activity of brain.
2. Describe the working principle of colorimeter.
3. State the properties of Laser.
4. State the use of respirators.
5. Classify the pacemakers and specify in which condition they are used.
6. State the applications of biotelemetry.
7. List the precautions to be taken while handling X-ray machines.

(5*6=30)

PART-C

(Answer one full question from each unit. Each question carries 15 marks.)

UNIT-I

III a). Explain the various types of electrodes used for ECG measurement .

10

b).Draw and explain the block diagram of EMG. 5

OR

IV a).With waveforms describe the resting potential and action potential of a cell. 8

b).What are the different types of brain waves. 7

UNIT-II

V a).Describe the different methods of blood cell counting. 8

b).explain the principle of operation of Nd-yag Laser with a neat diagram. 7

OR

VI a).With a neat figure explain blood gas analyzer . 8

b) Explain the application of Laser in medical field. 7

UNIT-III

VII a).State the need of defibrillators and explain the functions performed by it. 8

b).Explain the shortwave diathermy unit. 7

OR

VIII a).With diagram explain the working principle of hemodialysis machine . 8

b).Explain microwave diathermy unit. 7

UNIT-IV

IX a).Explain the working principle of CT scanner. 8

b).Draw and explain the block diagram of Biotelemetry system. 7

OR

X a).Describe the working principle of ultrasonic imaging system. 8

b).Explain macroshock and microshock. 7

